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# Plant–soil negative feedback explains vegetation dynamics and patterns at multiple scales

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Spatial patterns and self-organization of plants has been a subject of fascination because the underlying mechanisms have been hard to determine, raising different explanatory hypotheses. Plant–soil negative feedback (PSNF) – defined as the induction of negative conditions for conspecific establishment – has been widely studied in both field and laboratory conditions, and conceptually demonstrated by some modelling works. We present a mechanistic model, integrating individual plants inside an agent-based framework, to explore the effects of PSNF on the spatial and temporal dynamics of virtual populations and communities of plants of diverse growth forms. This endeavour led to the reproduction of well-known vegetation patterns observed at various scales, demonstrating for the first time a unified mechanism behind the spatial patterns of Janzen–Connell seedlings' distribution, ring formation, and the high species mobility in species-rich grasslands. These results support the ecological relevance of PSNF in the regulation of spatial organization and biodiversity dynamics in plant communities. More specifically, PSNF due to autoxicity seems most coherent with the spatio-temporal scale of dynamics displayed here.

Regular vegetation patterns such as fairy rings (Bonanomi et al. 2014), tiger bush bands (Leprun 1999) and stripes (White 1971) have fascinated plant ecologists for a long time (reviewed by Rietkerk et al. 2004). Vegetation patterns can be caused by the uneven feeding activity of insects, plant pathogens and grazers (Ponge et al. 2015), but also by the heterogeneous distribution of soil resources. Small scale below-ground heterogeneity in resource distribution is a common feature of several ecosystems, with co-occurrences in close proximity of nutrient poor and nutrient rich microsites reported in grassland (Jackson and Caldwell 1993), scrubland (Maestre et al. 2003) and forest (Kleb and Wilson 1997). At community scale, small-scale heterogeneity in the below-ground environment bears great importance because it can affect the performance of individual organisms and thus, the spatial and temporal distribution of plants.

Plants themselves can induce below-ground heterogeneity through several mechanisms: active excretion of organic compounds from roots (Neumann and Martinoia 2002), release of chemical compounds during organic matter decomposition (Chomel et al. 2014), modulation of soil microbial organisms and pathogens (Garbeva et al. 2004), local depletion of soil nutrients (Wedin and Tilman 1993), and water uptake (James et al. 2003). A specific effect of plant-induced soil heterogeneity is the so called plant–soil negative feedback (PSNF) (Bever 1994), i.e. a relative negative condition for the establishment, growth and reproduction of individuals induced in soil by conspecific plants. PSNF has been reported in different ecosystems including temperate and tropical forests (Streng et al. 1989, Mangan et al. 2010), coastal dunes (van der Putten et al. 1993), old fields and grassland (Blomqvist et al. 2000, Olff et al. 2000). The occurrence of PSNF has been related to species coexistence (Bever 1994), ecological succession (van der Putten et al. 1993, Kardol et al. 2006), invasion processes (Callaway et al. 2004), positive diversity–productivity relationships (Schnitzer et al. 2011) and, recently, to ecosystem restoration (Middleton and Bever 2012). Different, nonexclusive putative mechanisms have been proposed to explain PSNF: depletion of soil nutrients (Ehrenfeld et al. 2005), build-up of soil-borne pathogens (van der Putten et al. 1993), shifts in soil microbial communities (Klironomos 2002), and release of allelopathic autotoxic compounds from decomposing plant litter (Singh et al. 1999, Armstrong and Armstrong 2001), and accumulation of extracellular self-DNA (Mazzoleni et al. 2015, Cartenì et al. 2016).

Irrespectively of the underlying mechanisms, a large body of empirical evidence demonstrates that PSNF affects plants of different growth forms (reviewed by Kulmatiski and Kardol 2008, van der Putten et al. 2013), including annual (Bezemer et al. 2006) and perennial herbs (Kardol et al. 2007),

shrubs (Bonanomi et al. 2008) and trees (Packer and Clay 2000). In this regard, it should be expected that growth forms having specific propagation modes evolved different escape strategies from "home soil" (sensu Bever 1994) where PSNF builds up. For instance, annual plants can avoid "home soil" by random searching through seed dispersal for safe sites. Differently, perennial clonal plants can actively move away by selective proliferation of new ramets in suitable soil patches (Olff et al. 2000, Blundell and Peart 2004), while trees and shrubs can avoid PSNF localized under their canopy via seed dispersal (Packer and Clay 2000). Very recently, differences in PSNF depending on plant traits and functional groups have been revealed (Cortois et al. 2016). However, the effect of plant growth form and PSNF on spatial and temporal patterns in plant communities has not been investigated so far. Existing mathematical models explored the impact of PSNF on species coexistence (Bever 2003, Bonanomi et al. 2005a), invasion by exotic species (Eppinga et al. 2006), maintenance of diversity in tropical forest (Mangan et al. 2010), and to explain biodiversity gradients at global scale (Mazzoleni et al. 2010). An exception at population scale is the modelling exercise by Cartenì et al. (2012), which demonstrated that PSNF associated to autotoxicity alone can generate "ring" patterns in perennial, tussock forming plants. In such model, plant ramets escape negative feedback rising in the "home" soil (i.e. inside the tussock) by radiating away through vegetative growth, in agreement with empirical data. A more recent modeling exercise by Marasco et al. (2014) showed that PSNF can also be a driver of Turing pattern formation.

The abovementioned models based on PSNF, with few exceptions, mainly used non-spatial simulations and never investigated the interaction between plant growth forms and PSNF on the emergence of vegetation patterns. To overcome such methodological limitations and to provide a common modelling framework to simulate several PSNF-related phenomena, we present here a spatially explicit hybrid model that is capable of representing different plant growth forms at individual level and thereby observe the population spatial and temporal dynamics. The specific aim of this work was to investigate the link between PSNF, plant growth forms and the formation of vegetation pattern.

## Methods

### Model description

In this section we provide a short summary of the model description. The full description of the model is provided in the Supplementary material Appendix 1, in which we follow the seven-points overview–design concepts–details (ODD) model description protocol formulated by Grimm et al. (2006, 2010).

The model is based on a generic description of PSNF, in which an abstract intraspecific autotoxic process is assumed and can implicitly stand for most detrimental changes in biotic or abiotic soil conditions reported to generate PSNF (e.g. build-up of soil pathogens or of toxic compounds affecting conspecifics). Accordingly, 'toxic compounds' is an umbrella term that hereafter refers to the agents giving rise to PSNF.

The model implementation follows a hybrid modeling approach coupling different modeling paradigms (Vincenot et al. 2016b). More specifically, it relies on the system dynamics (SD) – individual-based (IB) hybrid modeling framework (Vincenot et al. 2011) to reproduce the spatio-temporal dynamics of a population of individual plants explicitly considering the PSNF exerted by modification of the soil environment resulting from litter decomposition (Cartenì et al. 2012, van der Putten et al. 2013). The hybrid solution is basically composed of two functionally related modules. The first module represents the individual plant growing according to a logistic curve (continuous process) up to a defined theoretical maximum mass which is modulated by plant–soil negative feedback. Once the reproductive age of the individual is reached, the plant starts to produce seeds, dispersed following a Weibull kernel (Weibull 1951), that can produce seedlings according to a specific germination probability (stochastic processes). It has to be noted that, due to computational limitations, seeds are not explicitly considered. Only germinated seedlings are produced as new agents if the conditions are met. The other processes considered are biomass turnover, which produces litter accumulated in the soil, and plant death, i.e. each individual has a stochastic lifetime centered around an average life expectancy parameter.

The second module simulates the dynamics in the soil of the autotoxic compounds, which are produced by plant biomass turnover and removed based on a constant decay rate (Mazzoleni et al. 2010). The levels of autotoxicity in the soil modulate two key plant processes: 1) the inhibition of the growth of individual plants and 2) the reduction in the germination probability of seeds. It is important to note that the effects of the toxic compounds are species-specific, i.e. affect only plants of the same species (Mazzoleni et al. 2015).

### Simulation experiments, parameters and analyses

The model was run to simulate the dynamics of plants of different functional types facing plant–soil negative feedback. In particular, we ran three sets of simulations to represent 1) woody plants that reproduce only by seed, 2) perennial herbaceous plants capable of vegetative propagation, and, 3) a plant community composed of short-lived herbaceous plants that reproduce only through seeds. All the model parameters set to represent different growth forms are reported in Table 1.

The first set of simulations was undetaken to study the effects of PSNF on the establishment of seedlings of woody plants and their cohorts' dynamics over the years. Each simulation was initialized with a single perennial plant in the middle of the spatial domain that grew according to a logistic curve up to a maximum biomass of $4 \times 10^7$ g. After reaching its reproductive age (≈15 years), the plant was set to produce a single seed dispersal event each year with a mean dispersal distance of 1 m (for the specifics on plant growth and seed dispersal processes see Supplementary material Appendix 1). Six simulations were performed using different levels of establishment sensitivity to PSNF (0%, 55%, 65%, 80%, 100% and 200%; the value of 100% refers to $\gamma = 34$, which is the value used in all other simulations). The spatial and age distribution of the seedlings recruitment curves as



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well as the size of the mother plant's crown were recorded in relation to negative feedback intensity.

In a second set of simulations, the model was parameterized to represent a perennial clonal plant with tussock structure such as the grass *Ampelodesmos mauritanica* (Incerti et al. 2013). Each of the ramets was represented by an agent with a maximum biomass of 400 g and a reproductive age of 180 days. To reproduce vegetative growth, the mean dispersal distance was set to 0.06 m. All simulations started with a patch of plants located in the center of the spatial domain. The spatial and temporal dynamics of the ramets were then recorded under different conditions of negative feedback intensity. Specifically, the first simulation was performed assuming no PSNF, whereas the second and third simulations included the effects of PSNF with different toxic decay rates ($k = 0.002$ and $k = 0.00001$). A last simulation at high PSNF intensity ($k = 0.00001$) was initialized with two separate patches of ramets of the same species to study the dynamics of coalescence between two rings.

In the third set of simulations the model was run simulating a community of 44 different species in a $10 \times 10$ m plot. We simulated a plant community composed of small (max biomass of 400 g), short lived plant species (average life span of 365 days) (Kull and Zobel 1991), with an average seed dispersal distance of 0.1 m. For simplicity, all the simulated species had the same characteristics (i.e. parameter values), but they only suffered the effects of their own species-specific PSNF. Simulations were initialized with 100 plants of each species randomly distributed over the plot. The simulation ran for 55 years in order to let the plants mix and cover the entire plot. We then analyzed the impacts of PSNF on species diversity at different spatial scales (0.001 $m^2$, 0.01 $m^2$, 0.25 $m^2$, 2.25$m^2$ quadrats) in the following 77 years.

### Data deposition

For demonstration purpose, a simplified version of the model presented in this paper is available online at <http://christian.vincenot.biz/models.html>. The full model has been stored at Dryad Digital Repository: <http://dx.doi.org/10.5061/dryad.6t8p0> (Vincenot et al. 2017).)

## Results

### PSNF on woody plants

In the case of seed dispersal by a single woody shrub or tree, the number of seeds on the ground followed the dispersal distribution with a density decreasing with distance from the rooting point. On the other hand, the accumulation of toxic compounds produced by litter decomposition was located below the crown of the mother plant, also with an intensity that decreased with distance from the trunk. Figure 1A shows that the number of established seedlings in presence of PSNF resembles the well-known Janzen–Connell recruitment distribution (Janzen 1970). Under the crown of the woody plant, notwithstanding the highest seed density, a zone was observed within which seedlings were totally absent. Further away, the number of seedlings progressively increased until it followed the shape of the seed dispersal curve out of the mother crown due to the absence of toxic compounds.

Table 1. List of model parameters and their values.

| | | | Growth form | | |
|---|---|---|---|---|---|
| Symbol | Description | Units | Woody plant | Clonal, perennial herb | Annual herb |
| – | Number of simulated species | – | 1 | 1 | 44 |
| – | Width of spatial matrix | pixels | 100 | 50 | 210 |
| $\Delta X$ | Pixel width | m | 0.25 | 0.04 | 0.0333 |
| $dt_{IBM}$ | Temporal integration step | $d^{-1}$ | 1 | 1 | 1 |
| – | Random initial spatial distribution | – | no | no | yes |
| – | Seed dispersal strategy | – | once per year | continuous | continuous |
| – | Initial number of plants | – | 1 | 10 | 100 |
| $B$ | Plant biomass (state variable initial value) | g | 1 | 0.5 | 0.1 |
| $g$ | Biomass growth rate | $d^{-1}$ | 0.502 | 0.519 | 0.519 |
| $d$ | Biomass death rate | $d^{-1}$ | 0.5 | 0.5 | 0.5 |
| $K$ | Theoretical maximum mass | g | $4 \times 10^7$ | $4 \times 10^2$ | $4 \times 10^2$ |
| $D$ | Plant biomass density | g $m^{-2}$ | 2000 | 2000 | 2000 |
| $K_c$ | Community carrying capacity | plant $m^{-2}$ | – | 3000 | 3000 |
| $L$ | Average plant lifespan | d | 365000 | 365 | 365 |
| $R$ | Reproductive age | d | 5500 | 180 | 90 |
| $T$ | Toxic compounds density (state variable initial value) | g $m^{-2}$ | 0 | 0 | 0 |
| $c$ | Proportion of toxic per biomass | $m^{-2}$ | $3 \times 10^{-7}$ | $1 \times 10^{-4}$ | $1 \times 10^{-4}$ |
| $s$ | Toxic sensitivity | $m^2$ $g^{-1}$ $d^{-1}$ | $1 \times 10^{-4}$ | $3 \times 10^{-4}$ | [0, $1.5 \times 10^{-3}$, $2 \times 10^{-4}$] |
| $k$ | Toxic decay rate | $d^{-1}$ | $5 \times 10^{-5}$ | [$2 \times 10^{-3}$, $1 \times 10^{-5}$] | $1 \times 10^{-3}$ |
| $S_0$ | Seed biomass | g | 1 | 0.5 | 0.1 |
| $P$ | Biomass-to-seed ratio | – | $1 \times 10^{-4}$ | $3 \times 10^{-3}$ | $4 \times 10^{-3}$ |
| $\mu$ | Dispersal kernel mean distance | m | 1 | 0.06 | 0.1 |
| $\sigma$ | Dispersal kernel shape | – | 1 | 1 | 1 |
| $\alpha$ | Base establishment probability | – | 0.6 | 0.6 | 0.6 |
| $\beta$ | Establishment sensitivity to toxicity parameter | – | $1 \times 10^{-5}$ | $1 \times 10^{-5}$ | $1 \times 10^{-5}$ |
| $\gamma$ | Establishment sensitivity to toxicity parameter | – | [0 68] | 34 | 34 |



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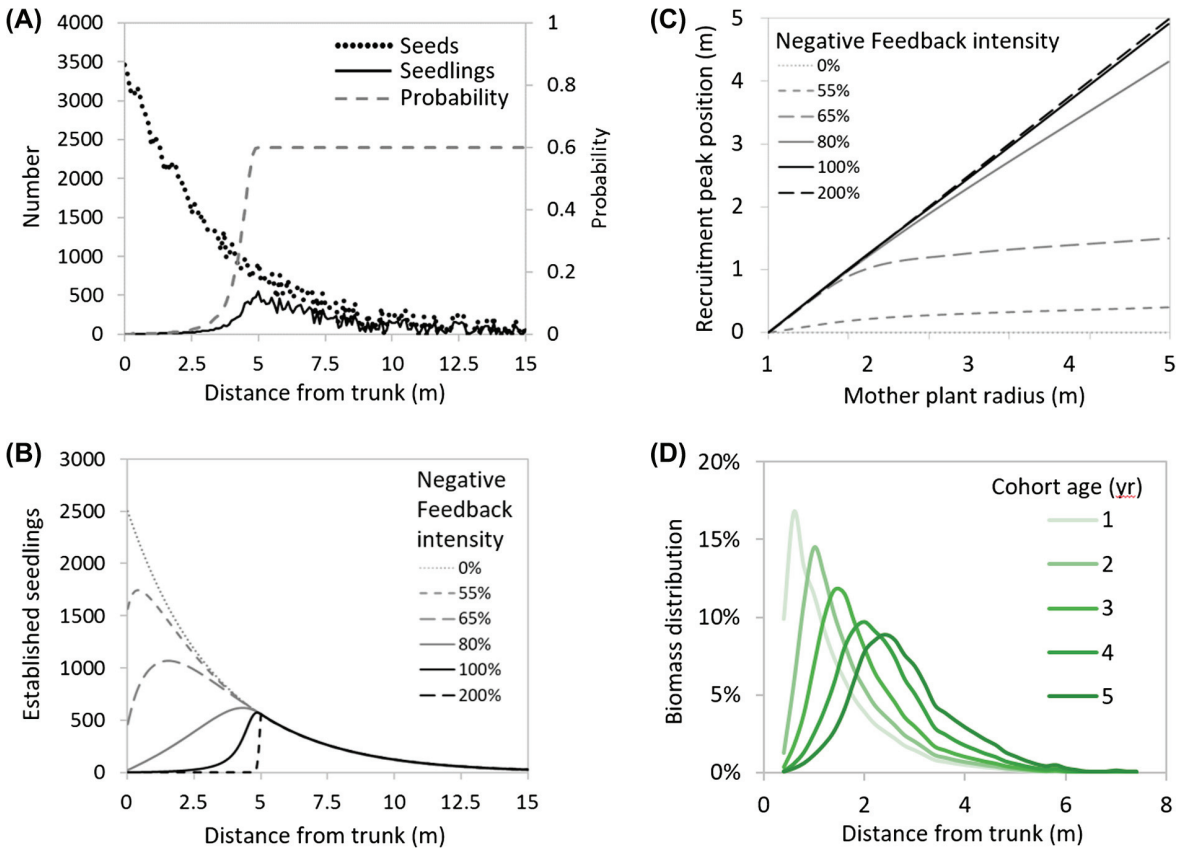


Figure 1. Effects of PSNF on seedling recruitment. (A-B) The simulations were initialized with a single perennial plant in the middle of the spatial domain that grew according to a logistic curve up to its maximum biomass. Results are shown after 22 years of simulation (i.e. 8200 steps). (A) Number of seeds produced (black dots), of established seedlings (black lines), and probability of establishment (dashed grey line). These together produce the well-known Janzen–Connell recruitment seedling distribution. The establishment sensitivity to toxic compounds γ for this simulation was set to 34. (B) Seedling recruitment curves at six levels of PSNF (expressed as percentages of the value of γ used in the previous simulation). (C) Position of the recruitment peak in relation to the radius of the mother plant's crown under different PSNF intensity. This plot was obtained by recording, for predefined values of PSNF (i.e. 0%, 55%, 65%, 80%, 100%, 200%), the recruitment peak position for different values of the mother plant's radius (from 1 to 5 m). At maximal PSNF (black continuous line) the peak of the recruitment curve is always at the border of the crown, whereas without PSNF (grey dotted line), the peak is always at the center of the mother plant where seed distribution is maximum. The other lines show the results for intermediary values of PSNF. (D) This simulation was initialized with a single perennial plant in the middle of the spatial domain, which, after reaching its reproductive age (15 years), produced one dispersal event each year. The plot shows the seedlings' biomass distribution curves (%) in the first 5 years after the first seed dispersal event. Other parameter values for these simulations are reported in Table 1 (Woody plant).

Figure 1B demonstrates the effects of different PSNF intensities on seedling establishment. Each line represents the average number of established seedlings relative to the distance from the trunk of the mother plant. The black line (100% intensity) simply reproduces the recruitment curve shown in Fig. 1A. Without PSNF (0%) the recruitment curve had the same shape as the seed dispersal curve, meaning that a constant proportion of seeds germinated at each given position. As the intensity of the PSNF increased, however, the peak of the recruitment curve lowered and progressively shifted towards the edge of the crown. In the extreme case of highest PSNF, no seedlings were present under the crown and establishment began right past the shadow of the mother plant where no toxic compounds were found in the soil. Figure 1C shows the position of the recruitment peak at different levels of PSNF relative to the size of the crown of the mother plant. In particular, if there was no PSNF effect, the peak was always positioned near the rooting point, thereby mirroring the seed dispersal kernel. On the contrary, if the PSNF effect was maximum, the peak was always located right outside the crown shadow. It is interesting to note that for intermediate values of PSNF the recruitment peak was always under the mother plant, but the position of the peak did not increase linearly with the radius of the crown, i.e. the bigger the crown the more the recruitment peak would shift towards the inside. PSNF also affected the spatial position of the peak of seedling recruitment as seedlings age in time (Fig. 1D). In presence of PSNF, we found a progressive shift of the recruitment curve outwards from the rooting point of the parent plant. In fact, the biomass peak of successive cohorts with age of 1, 2, 3, 4 and 5 years peaked further and further away from the parent plant.

## PSNF on herbaceous, clonal plants

The spread of the clonal plant occurred, in absence of other plants, in a roughly circular regular fashion. In the absence of PSNF (Fig. 2, left column), this resulted in the generation of



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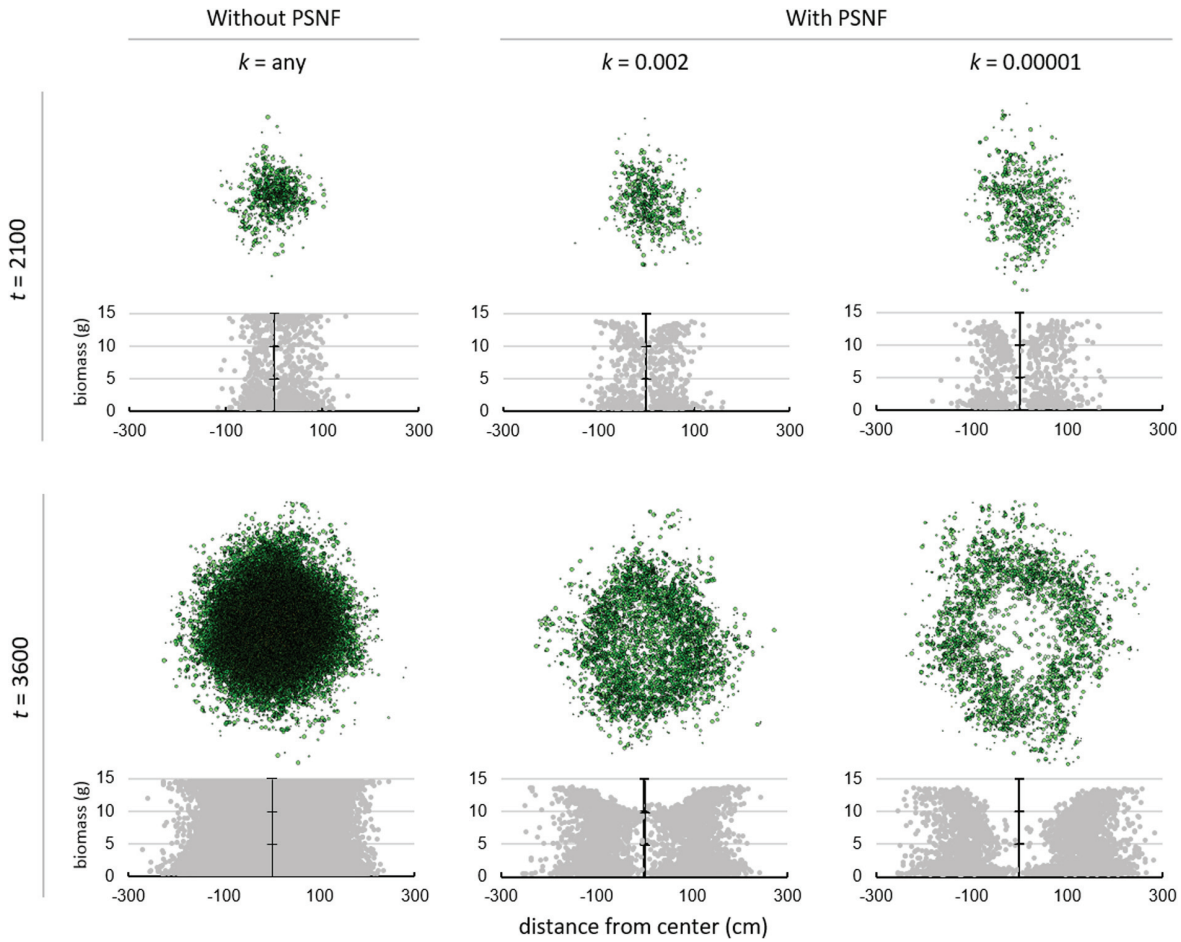


Figure 2. Effects of PSNF on ring formation by a clonal plant. The model was parameterized to reproduce a perennial clonal herb. All simulations started with a patch of ramets in the center of the domain and reproduced the propagation of a tussock with three different levels of PSNF. In this case, to represent PSNF intensities, decay rates of toxicity were changed (k = 0.002 and k = 0.00001) and each column represents a single simulation in such conditions (PSNF intensity increasing from left to right). Each subpanel shows a 2D map depicting the position of the ramets and a plot with the size and position from the centre of each ramet. Other parameter values are reported in Table 1 (Clonal, perennial herb).

a spot with the highest biomass in the center of the patch. If PSNF were introduced, the accumulation of toxicity in the center of the patch limited inward recolonization (Fig. 2, center and right column). The tiller population of the vegetative plant then developed into a ring-shaped spatial pattern, the "sharpness" of which depended on toxicity decay rate $k$. At high values of litter decay rate ($k = 0.002$), the inner part of the ring was still populated by plants but with lower biomass (Fig. 2, central column), whereas a lower decay rate ($k = 0.00001$) would allow only for the subsistence of a very low density in the inner part of the clone (Fig. 2, right column).

When the model was initialized with several plants and was left running, a phenomenon of temporal instability of the spatial organization of plants gave rise to the formation of arc-shaped biomass fronts travelling across space as waves (Supplementary material Appendix 2), consistently with previous works on the formation of ring patterns (Cartenì et al. 2012). In a further simulation, we also investigated the interaction between two plants of the same species. Figure 3 shows the time evolution of a simulation set up with two patches starting from different regions of the two-dimensional spatial domain. The two plants initially formed two distinct rings. Then, the two plants merged into a single vegetation patch in which the ramets of the two individuals were still spatially separated with a central ring still visible (Fig. 3, nine years). However, after 12 years of interaction from the beginning of the simulation, the plants showed the first signs of coalescence as the center of the ring began to be recolonized by both individuals. With time, the two clonal plants completely merged together and began producing a homogeneous cover of the soil with the disappearance of the initial ring (Fig. 3, 15 years).

## PSNF on short lived plants

Simulations performed on a community composed of 44 species showed that the plant community biomass as well as the total number of plants remained remarkably constant. The number of species found at each sampling size (0.00 1m$^2$, 0.01 m$^2$, 0.25 m$^2$, 2.25 m$^2$) remained roughly constant over time (Fig. 4A, dashed lines). However, in spite of the stability in species richness at different plots scales, notable species mobility was observed, with plants continuously occupying and then releasing in time the previously occupied soil patches. In fact, while no significant temporal change was visible in average species diversity, average



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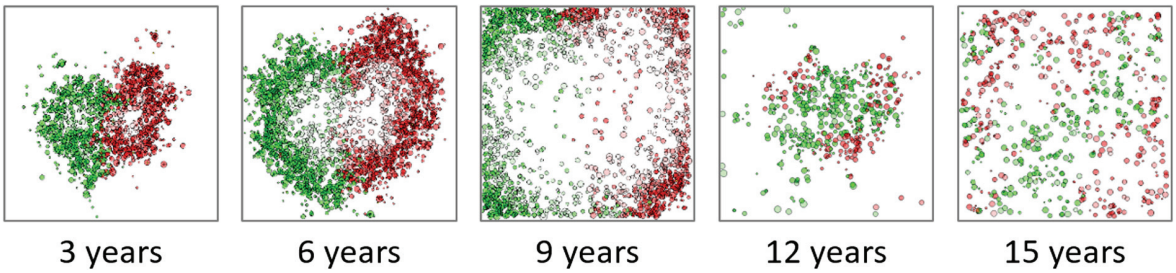


Figure 3. Effects of PSNF on two clonal plant individuals. The panels show 2D maps of the evolution in time of two ring-forming individuals of the same species (red and green). The simulation was initialized with two separate patches of ramets of the same species to study the dynamics of coalescence between two rings in presence of high levels of PSNF (k = 0.00001). Other parameter values are reported in Table 1 (Clonal, perennial herb).

cumulative diversity kept increasing in a saturation manner until reaching the total species pool of the community (Fig. 4A, solid lines).

The continuous increase in cumulated species diversity was likely due to the effect of PSNF, which disfavors individuals of a species to germinate and grow in places previously occupied by plants of the same species. In order to test this hypothesis, a new set of simulations was performed with different levels of PSNF (Fig. 4B). The results for the smallest sampling plot (0.001 $m^2$) showed that in presence of high PSNF the system experienced a very fast rotation and the cumulative number of species increased faster compared to simulations with low levels of PSNF. Moreover, without PSNF, the accumulation of plant species at the small spatial scale was very slow, with plant species achieving large local dominance.

## Discussion

This hybrid individual-based model relies on a broad definition of PSNF that translates in the model in a theoretical intraspecific autotoxic process, which conveniently embraces any process that has been shown to induce environmental changes in the soil that are detrimental to conspecifics. The presented results demonstrate that the interaction between PSNF and growth forms with different life spans and dispersal strategies may explain the formation of different spatial and temporal patterns at community scale. To avoid "home" soil, perennial plants with as single rooting point produce the classic Janzen–Connell recruitment distribution (Janzen 1970), clonal spreading plants form wave-like patterns and rings (Bonanomi et al. 2014), while short lived plants show a rapid and continuous turnover at small spatial scales that, at the same time, results in a stable plant assemblage at community scale. The latter specific spatial-temporal pattern well resembles the so called "carousel" model reported for species-rich grasslands (van der Maarel and Sykes 1993).

### PSNF as driver of the Janzen–Connell distribution

In the present model, perennial plants with a single rooting point subject to local PSNF under their own canopy exhibit a distance-dependent inhibition of seedling recruitment. Seedling establishment is reduced or even completely absent under the canopy as the intensity of PSNF increases. At high

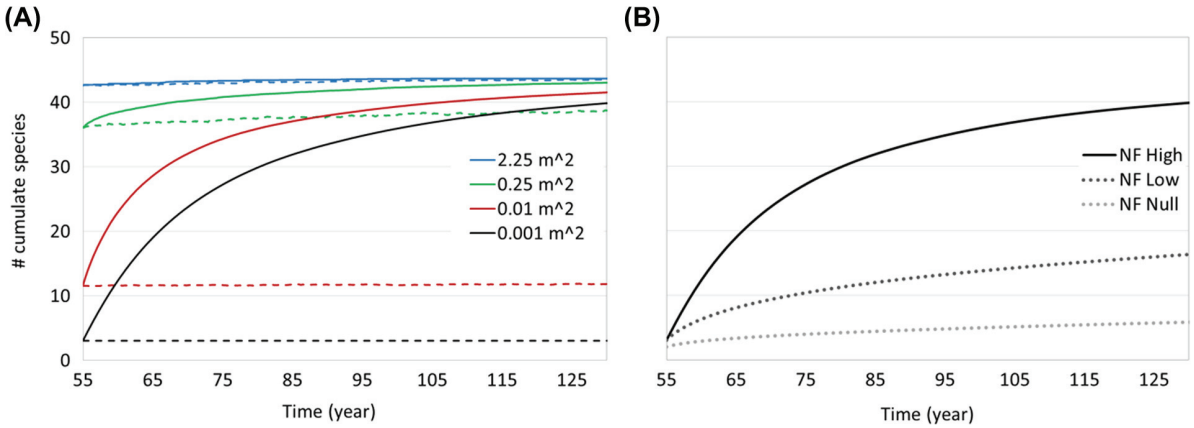


Figure 4. Effects of PSNF on a short lived plant community. The model was run to simulate a community of 44 different herbaceous species in a 10×10 m plot. All species had the same characteristics (i.e. parameters), but were only affected by PSNF produced by individuals of the same species. Plants were left 55 years to disperse and mix within the community. Diversity measurements started only after this warm-up period. (A) Dotted lines represent the average number of species recorded at each of the four sampling plots (0.001 $m^2$, 0.01 $m^2$, 0.25 $m^2$, 2.25 $m^2$), while solid lines represent the cumulated number of recorded species. (B) Effects of three different levels of PSNF on the cumulative number of species recorded in the smallest of the sampling plots (0.001 $m^2$). Note that the solid black line in this plot corresponds to the solid black line in the left plot. Different levels of PSNF were assumed by changing the value of s (s = 0.0002, dotted black line; s = 0, dotted grey line). Other parameter values are reported in Table 1 (Annual herb).



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intensity, PSNF effects can outweigh the higher seed fall under plant canopy, thus producing the well know Janzen–Connell recruitment distribution (Janzen 1970). Available empirical evidence demonstrated that PSNFs are implicated in the appearance of the Janzen–Connell pattern because of reduced success of conspecifics in the proximity of adult individuals. This phenomenon has been reported for perennial herbs (Petermann et al. 2008), shrubs (Bonanomi et al. 2008) and, most commonly, for trees in temperate and tropical forests (Mangan et al. 2010, Packer and Clay 2000).

Nathan and Casagrandi (2004) developed the first mathematical model reproducing the Janzen–Connell distribution. The model was based on localized activity of natural enemies with distance and density-dependent seed predation. It generated all possible recruitment patterns, including the Janzen–Connell distribution, in relation to predation intensity and shape of the dispersal kernel. Our individual-based model demonstrates that PSNF can also produce the Janzen–Connell distribution in plant recruitment if PSNF intensity is sufficient to overpower the seed dispersal kernel. Noteworthy is that our simulations show that the recruitment distribution not only resembles the Janzen–Connell distribution with the peak at intermediate distances from the rooting point, but that such peak shifts outwards from the seed source during seedling development. This pattern has been recorded in field conditions (Augspurger 1983, Gavin and Peart 1997, Blundell and Peart 2004, Arrieta and Suarez 2005, Norghauer et al. 2006, Bonanomi et al. 2008), and in our simulations can be related to the continuous radial growth of the plant canopy and the effect of PNSF on underlying soil. This result highlights a further outcome emerging from our simulations: a strong relationship between plant size and the position of the recruitment curve peak. In all simulations, the recruitment curve peaks were within the canopy or just outside. In agreement with field evidence, the position of the recruitment peak can range from few centimeters to several meters from the rooting points. Several studies showed that for herbaceous plants the recruitment peak occurs only few centimeters from the canopy edge (Mccanny and Cavers 1987, Kjellsson 1991, Bonanomi et al. 2005b). Bonanomi et al. (2008) reported for the small shrub *Medicago marina* that seedling abundance peaks at 30–50 cm from the rooting point, just few centimeters outside the canopy edge. Moreover, previous studies from tropical forests showed that population recruitment curves peaked at 1 or 2 m from the rooting point of small dominated shrubs (Condit et al. 1994), while it occurred at several meters, in some cases more than 10 or even 20 m from the bole of large, dominant trees (Augspurger 1984, Clark and Clark 1984, Cintra and Horna 1997). The strong correlation between plant size and inhibition zone occurs because litter accumulates under the canopy of the source plant, and PSNF strength decreases monotonically with distance from the rooting point (Staelens et al. 2003). Accumulated litter can then induce negative conditions through the release of autotoxic compounds and/or affecting the soil pathogen and microbial communities. As a result, the highest inhibition occurs close to the rooting point and decreases in a distance-dependent way. However, it must be noted that the spatial scale of many observed Janzen–Connell distributions seems more consistent with diffusion processes of inhibitory compounds related to litter accumulation (Mazzoleni et al. 2015), while less compatible with the mobility range of pathogens which is plausibly larger than the observed patterns. For example, the inhibition zone of seedlings' recruitment of *Prunus serotina* attributed to *Pythium* spp. by Packer and Clay (2000) extends for a few meters, but the invasion speed of this pathogen is approximately 2–3 cm day$^{-1}$ (Nelson 2004).

## PSNFs and 'ring' formation in clonal plants

In a recent review, Bonanomi et al. (2014) reported that herbs, shrubs and even trees with clonal propagation, during their ontogenetic cycles, produce clones in a "ring" shape that progressively degenerate in the older inner area, thus producing a dieback central zone. This vegetation pattern has been called fairy rings (Hitchcock 1935), rings (Watt 1947), hollow crowns (Strickland 1983), central dieback (Adachi et al. 1996), and monk's tonsure-like gaps (Lewis et al. 2001). Noteworthy is that the majority of the "ring" shaped plants has been found during colonization of bare substrates or in environments with discontinuous plant cover such as deserts (Danin 1996, Sheffer et al. 2007), primary succession over bare substrates such as lava flow (Adachi et al. 1996), harvested peat lands (Lanta et al. 2008), salt marshes (Castellanos et al. 1994), and sand dunes (Pemadasa 1981). The results of our model are consistent with these empirical findings because plants form rings only during the early simulation phase when they colonize the empty simulation arena. Interestingly, when different clones come in contact, the ring shape disappears because of the coalescence between different clones. This is consistent with the rarity of "ring" shaped plants in ecosystems with continuous vegetation (Bonanomi et al. 2014). Further studies are required to investigate this process in real plant communities.

At least four different mechanisms have been proposed to explain "ring" formation in clonal plants, namely 1) co-occurrence of architectural constraints and natural senescence of inner ramets (Adachi et al. 1996), 2) induction by physical disturbance and stresses such as fire, drought or overgrazing (Strickland 1983), 3) resource depletion by intra- and inter-specific competition inside the ring (Sheffer et al. 2007), and 4) onset of PSNFs in the inner zone of the clone (Bonanomi et al. 2005). Our model demonstrates that PSNFs, alone, can putatively explain "ring" formation in clonal plants irrespectively of resource and water depletion in the inner area. This occurs because the individual tiller in the inner area of the clones are completely or partially inhibited by the negative conditions (i.e. autotoxic compounds or pathogens) accumulated in the 'home' soil. This phenomenon seems well fitting with the concept of accumulation in the central part of the tussocks of inhibitory self-DNA from the plant decay processes (Mazzoleni et al. 2015). Otherwise, it is difficult to hypothesize a role of pathogens as causal mechanism of PSNF at the spatial scale of an individual plant ring. In this regard, PSNF due to autotoxicity seems the most plausible mechanism capable of explaining both the formation of a complete die-back as well as a partial weakening inner area (Bonanomi and Allegrezza 2004, Otfinowski et al. 2016), in relation to PSNF intensity and dynamics. For instance, the architectural hypothesis predicts



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that recolonization of the senescent inner area is precluded to new ramets due to two main physical constraints: their preferentially centrifugal direction of growth, and the presence of remains of senesced ramets accumulated in the inner area of the clone (Feldman et al. 2007). This, again, logically seems to reflect a sensing of negative conditions from the roots rather than a pathogen effect.

Several models in the last decades investigated the mechanisms underlying ring formation in clonal plants (for a complete review see Bonanomi et al. 2014). The first model to explicitly test the emergence of rings of the clonal plant *Solidago altissima* was a stochastic simulation model presented by Cain et al. (1991), showing that factors other than rhizomatous growth patterns could be responsible for this phenomenon, such as resource depletion, clonal senescence and competition. Later on, other authors investigated the effects of water limitation as a putative mechanism for ring formation. Meron et al. (2007) developed a partial differential equation (PDE) model to study the interactions between biomass growth and water availability in semi-arid ecosystems. The authors showed how plants with long rooting systems develop spot-like patches due to the depletion of water in their surroundings, thus preventing further expansion. On the other hand, plants with phalanx growth strategy with short roots tend to form ring patterns because the local water depletion forces them to grow laterally, creating water stress conditions in the middle of the patch that induces central dieback. More recently, Cartenì et al. (2012) proposed a simple PDE model based on the concept of PSNF by litter autotoxicity (Mazzoleni et al. 2010), providing a putative mechanism for the formation of ring patterns where water is not a limiting factor. The model presented here extends the work by Cartenì et al. (2012), showing that similar results can be achieved also with a system dynamics (SD) – individual-based (IB) hybrid approach. Such approach brings several advantages such as the possibility to define individual growth form related characteristics (e.g. dispersal/propagation strategy) and to observe community scale results (e.g. age and biomass distribution of the individuals). Moreover, this approach could be extended to also explicitly consider other mechanisms such as architectural growth constraints and water limitation (as in Vincenot et al. 2016a).

## PSNF and species turnover of short lived plants

More than twenty years ago, van der Maarel and Sykes (1993) reported a high species mobility in species-rich grasslands. According to their field observations, each of the coexisting plant species continuously changed place within the target community area, thus producing counterintuitive temporal variations in species richness (i.e. stable scale-independent species diversity, but cumulative values increasing in time at every scale). Because of the continuous and apparently random species turnover occurring at small scales that contrasts with the almost constant species richness observed at community level, the authors called such spatial-temporal pattern the 'carousel' model. Following studies reported a similar high species turnover in other grassland systems (Sykes et al. 1994, Klimes 1999, Palmer and Rusch 2001), but the underlying drivers are still not fully understood. Small scale, short-term heterogeneity in soil resources and water availability may be possible ecological factors involved in the 'carousel' effect, but no study was able to empirically demonstrate these effects. The hybrid individual-based model presented here was able to reproduce for the first time the 'carousel' model showing continuous species turnover at small spatial scales coupled with a species accumulation until reaching the community species pool. In simulations, coherently with observations made by van der Maarel and Sykes (1993), each microsite is first colonized and then becomes a niche for all of the species included in the community. Model simulations showed that PSNF intensity strongly affects the turnover of plant species and the whole dynamics of the plant community. In fact, the 'carousel' takes place only in presence of strong, highly species-specific PSNF. On the contrary, when PSNF is weak or absent each different species establishes high local dominance that is maintained over time, which results in negligible changes in cumulative species diversity.

A large and still growing body of empirical evidence demonstrates that short-lived, herbaceous species suffer strong PSNFs (reviewed by van der Putten et al. 1993, Kulmatiski and Kardol 2008). However, the majority of such evidence has been found in laboratory, greenhouse and, in few cases, in common garden experiments (Kulmatiski and Kardol 2008). In this respect, a parallelism between natural grasslands and agroecosystems can be noted: in the first case plants move away from previously occupied soil patches through dispersal, while in agro-ecosystems farmers overcome the PSNF, also called 'soil sickness' or "replant disease problem" (Miller 1996, Huang et al. 2013) through the ancient but effective agronomic practice of crop rotation. Indeed, all the evidence presented suggests a potential link between PSNF and the small scale, short-term species mobility observed in grasslands. Future experimental studies, however, are required to demonstrate the existence of a link between PSNF and the 'carousel' effect, which seems to further reduce the spatial resolution of the applicability of the PSNF model.

## Modelling prospects

Further research on PSNF is needed as much remains to be discovered on the involved mechanism(s) as well as the effects of their interactions with other biological processes and environment conditions. Modelling seems like a promising approach to disentangle this complexity and study efficiently factors that are challenging to control in field experiments. The presented hybrid model – through its simplicity as well as its technical capability to render spatially plant individuals concurrently to continuous processes occurring in plants and soil – represents a framework to support future scientific undertakings. As shown here, it enables theoretical investigations on spatial and temporal phenomena observed at various scales as well as their subsequent linkage and comparison with empirical data. The generic specifications chosen to simulate plants – while not capturing many biological processes that underlie PSNF – simplify further hypothesis testing on plant traits.

## Conclusion

The existence of negative feedback from the soil to individual plants has been verified and quantified through numerous



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laboratory experiments in recent years. This study demonstrated how, when integrating PNSF at the individual-scale in a simple simulation model, it explained patterns observed in plants of various growth forms (woody, clonal perennial herbs, and annual herbs) at different scales (from individuals to communities). This exercise further supports the idea that PSNF – in particular due to autotoxicity, which involves diffusion processes that are compatible in scale with the patterns observed here – can be the putative mechanism regulating patterns observed in the spatial arrangement and biodiversity dynamics of plant communities, with obvious implications both in ecology and agricultural sciences.

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Supplementary material (available online as Appendix oik-04149 at <www.oiksjournal.org/appendix/oik-04149>). Appendix 1. Full model description following the ODD format. Appendix 2. Video showing the 'travelling waves' vegetation pattern.

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